The Reversible Flattening: Algebraic Representation as a Recoverable Compression of Relational Structure

A homomorphism criterion for when scalars should be returned to geometry

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Parent of *The Geometry Engine* / *When a Number Lies* — SIP License v1.1 — Seed 20260423

Abstract

We advance a single mathematical thesis with a precise, testable criterion, and characterize the boundary it runs into. Much of arithmetic and algebra is a *flattening*: a relational object (a factorization, a divisor lattice, a dominance relation) is projected onto a scalar or a symbol string because, historically, symbols on paper were the computable medium. We give the exact condition under which that flattening is *reversible* — recoverable losslessly to a relational form one can compute in directly — and show it is the condition that the operation producing the scalar be a *homomorphism* from the object's structure into a structure the relational representation stores. The boundary of reversibility (the "seam") is not a void: we characterize it precisely as *convolution* — the operation that adds indices while multiplying values — and verify that the additive structure it was thought to lack is real but *cardinal* (the partition function, the size of the cloud) rather than *unique* (a factorization, a point), and is encoded multiplicatively by its generating function (Euler's product, verified). Two kinds of structure — unique and cardinal — meet in convolution, which is crossable for density and uncrossable for identity; this single fact organizes the explicit formula, the difficulty of fully homomorphic encryption, and the limits below. Under this criterion, three regimes appear and are each verified. (1) The *homomorphic* regime — multiplication, gcd, lcm, powers, and every multiplicative arithmetic function — is reversible: by the Fundamental Theorem of Arithmetic, $(\mathbb{Z}_{>0}, \times)$ is isomorphic to $(\bigoplus \mathbb{N}, +)$ on exponent vectors, and we verify exact computation across seven applied-mathematics domains, 14/14 tests exact against ground truth, including objects of ~ 5000 digits that overflow the scalar form. (2) The *non-transitive* regime — dominance relations, social choice — admits *no* faithful scalar at all: a Condorcet cycle forces every total order to contradict the majority relation, so the relation is mandatory, not merely preferable, and the cyclicity persists at scale. (3) The *seam* — addition and ordering — is non-homomorphic on factor structure and therefore irreversible: it is the genuine, fundamental wall, proven by direct counterexample. The thesis is thus not "geometry is better than algebra" but the sharper claim that the algebraic flattening was a medium-driven compression, reversible exactly on the homomorphic and relational classes, and that modern computation lets us both perform and *verify* the inverse transposition. We keep the full negative ledger — two falsified search heuristics and four architecture-integration hypotheses falsified against live source — because the boundary, not the enthusiasm, is the result.

1 Thesis

Mathematics began with observation and construction — lengths, angles, areas — and was then abstracted into symbol and number because that form was mechanically manipulable by hand. This abstraction was an enormous gain and also a *compression*: it flattened relational content into coordinate- and digit-strings, and a great deal of later effort (managing precision, overflow, coordinate artifacts) is the cost of recovering relations the flattening discarded. The thesis of this paper is precise:

> The algebraic flattening of a relational object into a scalar is *losslessly reversible* if and only if the operation that produces the scalar is a homomorphism from the object's structure into a structure the relational representation stores directly. Where reversible, computing in the restored relational form is more faithful and frequently more tractable; where not, the scalar is native and the flattening is genuine.

This is a statement about *representation*, not about the superiority of one notation. It explains a historical contingency (symbols were the computable medium) and supplies a criterion that did not previously have a name, then verifies the criterion across applied mathematics.

2 The criterion

Definition 1 (Flattening and its inverse)**.** *A* flattening *of a structured object R is a map $\pi : R \to s$ to a scalar or symbol string. It is* reversible *if there is a representation $\rho(R)$ — the relational form — and a recovery $\rho(R) \to s$ that is exact, such that the operations of interest are computed on $\rho(R)$ without forming s.*

Criterion 1 (Homomorphism)**.** *The flattening of an operation $\star$ is reversible exactly when $\star$ is a homomorphism $(\text{objects}, \star) \to (\text{stored relational structure}, \oplus)$. Then $\rho(a \star b) = \rho(a) \oplus \rho(b)$ and the relational form computes $\star$ without forming the scalar.*

For integers under multiplication this is the Fundamental Theorem of Arithmetic: factorization is an isomorphism $(\mathbb{Z}_{>0}, \times) \cong (\bigoplus_p \mathbb{N}, +)$, so multiplication becomes vector addition of exponents, gcd and lcm become coordinatewise min and max, powers become scaling. We verify $\rho(ab) = \rho(a) \oplus \rho(b)$ directly. Addition fails the criterion: $\rho(a + b)$ is not a function of $\rho(a), \rho(b)$ in any structure-preserving way — $360 + 84 = 444$ and $363 + 81 = 444$ have identical sums from disjoint summand structures, and a unit change of input $(360 + 86 = 446)$ yields an unrelated factor structure. Addition is not a homomorphism on factor structure; its flattening is irreversible. This is the seam, now stated as the exact non-homomorphic boundary rather than as a list of hard cases.

3 Regime 1: the homomorphic class, verified exact

Every operation that is homomorphic into exponent-vector structure is reversible, and we computed each in the relational form, exact against ground truth (oracle: standard libraries; seed `20260423`).

domain	operations	verification
arithmetic functions	$d(n), \sigma(n), \sigma_2(n), \varphi(n), \mu(n)$	exact, 300 values each
gcd / lcm	min/max on exponents	exact, 300 pairs each
binomial / factorial	Legendre exponent formula	$\binom{20}{7}, \binom{100}{50}, \binom{500}{123}$ exact
modular products	homomorphism preserved under mod	concentrated chain mod $(2^{61}-1)$ exact
lattice index / SNF	product of elementary divisors	$\sim$1479-digit index, exact, no overflow
multinomial multiplicity	$N!/\prod n_i!$	$\sim$4978-digit W, exact
radical / perfect power	squarefree kernel, exponent gcd	exact, 200 values

Observation 1 (14/14 exact; the flattening visibly fails where the relation does not)**.** *All fourteen tests are exact. Two cases are decisive for the thesis: the multinomial W (*~ 4978 *digits) and the product of the first* 5000 *primes (*$\sim$ 20975 *digits) are computed exactly in relational form, while* forming the scalar *exceeded the host language's integer-to-string limit — the flattening hit a wall the relational representation did not. The headline tractability instance (companion FF06g/h) is an exact-optimal search for the integer maximizing d(n) that runs in exponent space to* $N = 10^{80}$*, verified-optimal against brute force at four anchors, sixty-two orders of magnitude past enumeration.*

4 Regime 2: the non-transitive class, where no scalar exists

The strongest form of the thesis is not lossiness but impossibility.

Theorem 1 (No faithful scalar under cycles)**.** *If the pairwise majority relation contains a cycle* $A \succ B \succ C \succ A$*, every total order contradicts at least one majority verdict. No scalar ranking represents the relation.*

Verification. Exhaustive over all orderings of the Condorcet instance: the minimum number of contradicted majority edges is one, not zero. Total orders are transitive; the majority relation is not; a transitive structure cannot realize a cycle. This is the social-choice form of the same obstruction by which box containment (a partial order) cannot encode rock-paper-scissors. □

Observation 2 (Structural, not small-sample)**.** *Measured cycle frequency rises with options (*$\sim$ $7\%, 43\%, 78\%, 99\%$ *for* $3, 5, 7, 10$ *candidates) and persists with electorate size (*$\sim 48\%$ *at* 1001 *voters, five candidates). The impossibility is a structural feature of aggregation. Here the relational representation (a directed dominance graph, integer and exact) is not an improvement on a scalar — it is the only faithful object.*

5 Regime 3: the seam, the irreversible wall

Addition and ordering are non-homomorphic on factor structure (§2) and so their flattening is irreversible: the sum genuinely destroys multiplicative structure. This is not a limitation of the representation; it is a fact about the operation, and it bounds the thesis from below. Every mixed object inherits it: a determinant is a *sum* of products (irreversible) even though each product is reversible; a characteristic polynomial's coefficients are elementary symmetric *sums* (irreversible) even when the eigenvalue *product* (the determinant's magnitude) is reversible. The criterion sorts these correctly without appeal to intuition: test whether the operation is a homomorphism into the stored structure.

6 The seam characterized: convolution (keystone)

The seam is not a void to be avoided; it has a precise structure, and naming it sharpens the entire thesis.

Theorem 2 (Two kinds of structure)**.** *The multiplicative side carries* unique *structure: by the Fundamental Theorem of Arithmetic a number's factorization is a single canonical object (a point). The additive side carries* cardinal *structure: a number's additive decompositions are non-unique (a cloud), but the* size *of that cloud is exact and deep — the partition function $p(n)$ (e.g. $p(100) = 190{,}569{,}292$). Addition is not structureless; its structure is the count, not an address.*

Theorem 3 (The seam is convolution, verified)**.** *The additive (cardinal) structure is encoded multiplicatively: Euler's identity $\sum_n p(n)x^n = \prod_{k\geq 1}(1 - x^k)^{-1}$ turns the entire additive cloud into one product (verified: the product's coefficients equal $p(n)$ for $n \leq 30$). The operation joining the two sides is* convolution*: multiplying generating functions convolves coefficients, $(a\,x^i)(b\,x^j) = ab\,x^{i+j}$, which* adds *the index and* multiplies *the value. Convolution is the native inhabitant of the seam — simultaneously additive in the index and multiplicative in the value (verified on coefficient sequences). The seam is therefore not impassable; it is crossed by generating functions, and the crossing operation is convolution.*

Observation 3 (Density, not identity)**.** *Convolution crosses the seam for* counting *and not for* identifying*: it yields how many, never which one. This single property organizes three facts at once. The Riemann explicit formula (companion work) is a convolution of primes against zeros, which is why the prime/zero relationship is statistical (a $+122\sigma$ density signal) and never an address. Fully homomorphic encryption is hard for the same reason the seam is real: addition and multiplication share no common homomorphic structure, so a single scheme must host both natively (lattice/LWE) and pay in noise — there is no cheap bridge, exactly as the seam predicts. And the additive wall of §2 is precisely the statement that convolution gives density without identity: the sum's factor structure is the convolution of the summands' structures, which determines a distribution, not a point.*

7 Two independent axes: order-dependence and structure-retention

A test of the conjecture "commutativity measures projected-away structure" falsified the strong form and yielded a sharper map. Non-commutativity measures retained *order-dependent* structure: scalar $+, \times$ are commutative and retain none (an index of 0); the cross product is anti-commutative, retaining exactly orientation (one bit, a structured sign flip); matrix products are fully non-commutative, retaining the entire arrangement. But commutativity does *not* imply absence of structure: gcd and lcm are commutative and retain the full common multiplicative structure (order-*independent* structure). Order-dependence and structure-retention are therefore *independent axes*. The shape engine lives in the commutative-and-structured cell — which is exactly why its canonical fingerprint is path-independent (commutativity) yet information-complete (structure). The corrected law: non-commutativity measures retained order-dependent structure only; order-independent structure is orthogonal to it.

8 Consequence: inherited representation as recoverable debt

The criterion gives "inherited technical debt" a precise mathematical reading. A representational layer built atop the algebraic flattening inherits the flattening's irreversibility wherever the underlying operation was non-homomorphic, and inherits a *recoverable* compression wherever it was homomorphic. The disciplined move, when seeking improvement, is therefore not only to build forward but to test whether a lower layer was homomorphically richer than the layer built on it — and, crucially, modern computation lets us *test* this per operation rather than assert it. The transposition back to relational form is a hypothesis checked by the homomorphism criterion and anchored against ground truth, not a faith. We make no claim beyond mathematics here: the criterion is the content.

9 Negative ledger (retained)

Falsifications are first-class. Two search heuristics were falsified en route to the verified-optimal $d(n)$ search (a greedy method, suboptimal; a non-admissible bound that pruned true optima). Four architecture-integration hypotheses were falsified against live source: the multiplicative-fit of an eigenvalue "charge" (the real charge composes additively, by the kernel's own specification); a geometric dual-engine router (the live constraints are ordinal thresholds and the live relation is transitive — no cycles); and trust and hash layers (additive and avalanche-destroyed, respectively). In each, a clean structural resemblance was the warning sign, and the source or the anchor was the arbiter. These are the criterion working: the homomorphism test predicts the misses as well as the hits.

10 Scope

The mathematics is classical (FTA, Legendre, the multiplicativity of σ, φ, μ, Smith normal form, Arrow/Condorcet). The contribution is the *criterion* — reversibility of a flattening equals homomorphism into the stored relational structure — and its verification across seven applied domains, the non-transitive impossibility class, and the additive wall. The thesis does not touch primality or prime-counting (building a prime's relational form is the factoring wall), does not bear on the Riemann Hypothesis (a categorical gap between finite multiplicative bookkeeping and the analytic continuation of an infinite object), and is not an arithmetic accelerator (a compiled big integer outpaces the relational form on raw throughput; the relational advantage is exactness, overflow avoidance, and faithful representation of the otherwise-unrepresentable). Within those bounds it is exact, reproducible, and bounded from both sides.

Parent of *The Geometry Engine* and *When a Number Lies* (the relational principle and limits). Reference engine, dominance engine, and the reproducible ledger `verify_ledger.py` (14/14 exact) accompany this paper. Seed `20260423`.